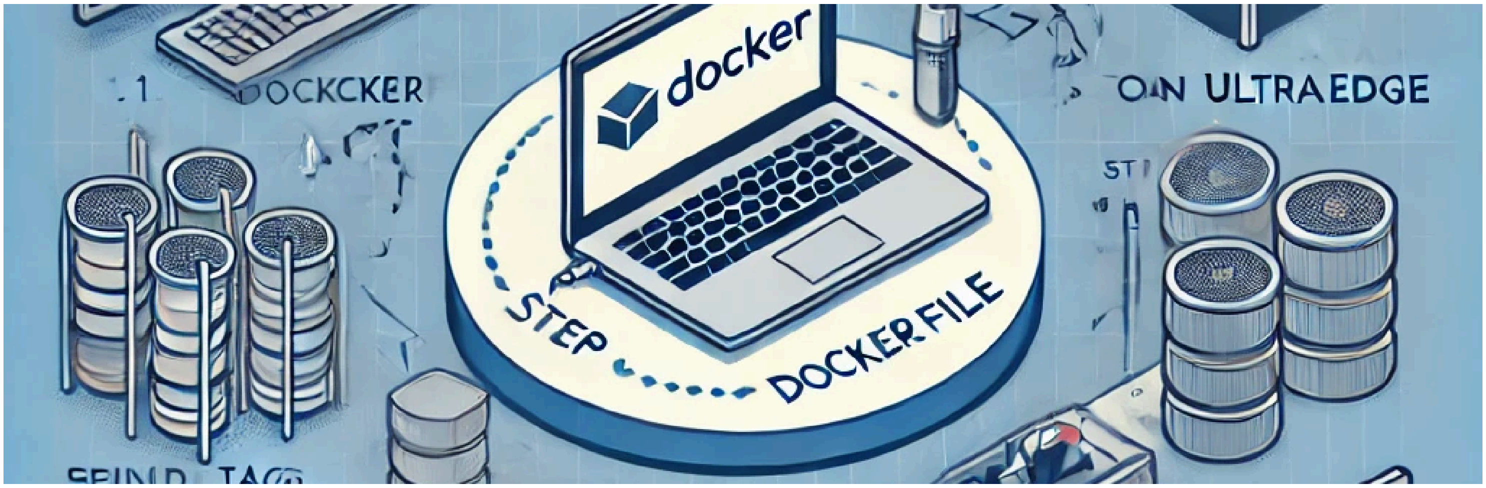


Send a Docker File to UltraEdge from IG-XL

Reference : RA 450

Introduction

This RA demonstrates how to send a Docker container from an IG-XL test program to an UltraEdge system. It includes:

- Loading the Docker container from IG-XL
- Running a Python script inside the container to transmit a poem through a FIFO
- Communicating between UltraEdge and VBT via FIFO

General Technical Info

Component	**Version**
Language	VBA & Python 3.11
Environment	Windows 10
IGXL	> 10.30.10
Excel	2016 64 bits
IDE	Visual Studio Code

Python Script & FIFO Communication

What the Script Does

- Sends a poem line-by-line via FIFO `/app/exec/fifo_to_TP`

- Appends `\0` to each message for proper delimiting
- Logs each transmission

Key Concept: What is a FIFO?

> A FIFO (First-In, First-Out) is a special file used for inter-process communication where data is read in the same order it's written.

Code Snippet

```
with open(FIFO_OUTPUT, 'w', encoding='utf-8') as output_stream:
    for line in POEM:
        output_stream.write(line + "\0")
        output_stream.flush()
        logging.info(f"Sent to FIFO: {line}")
```

Docker Container Setup

Dockerfile

```
FROM python:3.10-slim
ENV PYTHONUNBUFFERED=1
WORKDIR /app
COPY sampleapp.py /app/
CMD ["/app/sampleapp.py"]
```

- Base image: `python:3.10-slim`
- Environment setup: disables `.pyc` creation & enables logging
- Copies the Python script to `/app/`
- Sets the script to run on container start

Build & Export

```
docker build -t sampleapp .
docker image save sampleapp:latest | gzip > sampleapp.tar.gz
docker image rm sampleapp:latest
docker image list
```

Connecting from IG-XL to UltraEdge

Connect & Start Session

```
UE.Connect "10.100.xxx", yyyy
UE.SecureSession.Start
```

Transfer, Load & Run Docker

```
UE.WriteFile theexec.TestProgram.Path & "\" & "sampleapp.tar.gz", "sampleapp.tar.gz"
UE.LoadDockerImage "sampleapp", "sampleapp.tar.gz"
UE.CreateFifo "sampleapp" & FROMUE_FIFO
UE.Application("Demo").RunDockerContainer "sampleapp", "/app/sampleapp.py", "sampleapp", ""
```

Read FIFO Output

```
UE.ReadString ApplicationName & FROMUE_FIFO, data
Call theexec.AddOutput("UE: " & data, vbGreen, False)
```

Stop the Container

```
UE.Application(ApplicationName).Terminate
Call DebugMessage("Application " & ApplicationName & " termination")
```

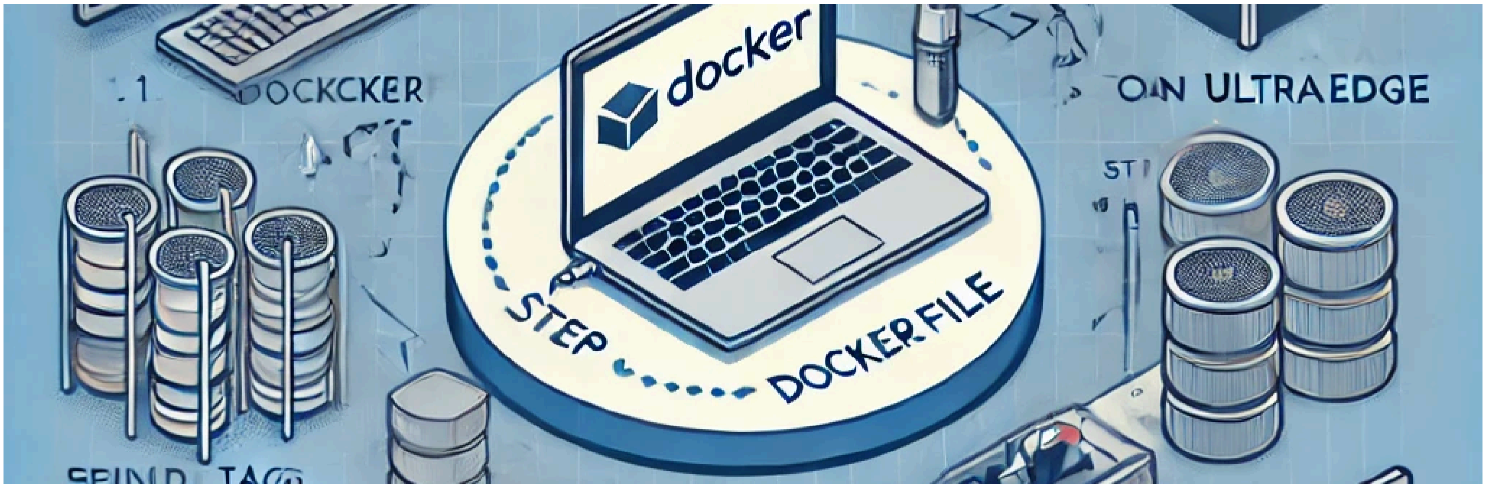
Source

File	Description
Docker Files	
Dockerfile	Configuration file for the Docker File Generation
makedocker.sh	Bash file for generating the Docker file based on previous file
sampleapp.py	Python script file with the FIFO in charge to communicate with the test program
Test Program	
sampleapp.tar.gz	Docker file generated required by the test program
uetrial.igxl	Test Program able to handle the dockerfile and communicate with the Python code

Summary

This RA provides a fully working example to:

- Package and export a Docker container
- Communicate via FIFO between Python and IG-XL
- Control container lifecycle via IG-XL VBA code



📄 **Description**	🔗 **Download Link**
Entire Solution	[The solution]
GIT Hub Link	[Git Hub]

More details about those source code here

This documentation is part of the Teradyne Archimedes Reference Architecture series.